

Patrick (Pengcheng) Jiang

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EDUCATION

University of Illinois at Urbana-Champaign

Ph.D. in Computer Science

Champaign, USA

Aug. 2024 – Present

- GPA: 4.00/4.00; Advisors: Prof. Jiawei Han (thesis) and Prof. Jimeng Sun
- Research interests: information retrieval, knowledge graphs, and LLMs for healthcare

M.S. in Computer Science

Aug. 2022 – May 2024

- GPA: 3.97/4.00; worked with Prof. Jiawei Han and Prof. Jimeng Sun

Waseda University

B.E. in Computer Science and Engineering

Tokyo, Japan

Sep. 2017 – Sep. 2021

- GPA: 3.85/4.00; worked with Prof. Shinichi Honiden and Prof. Kenji Tei

PREPRINTS

- [Preprint] **Pengcheng Jiang**, Jiacheng Lin, Lang Cao, Runchu Tian, SeongKu Kang, Zifeng Wang, Jimeng Sun, and Jiawei Han, “DeepRetrieval: Hacking Real Search Engines and Retrievers with Large Language Models via Reinforcement Learning”. [\[pdf\]](#)[\[code\]](#)
- [Preprint] **Pengcheng Jiang**, Lang Cao, Ruike Zhu, Minhao Jiang, Yunyi Zhang, Jimeng Sun, and Jiawei Han, “RAS: Retrieval-And-Structuring for Knowledge-Intensive LLM Generation”. [\[pdf\]](#)[\[code\]](#)

PUBLISHED WORKS

- [ICLR’25] **Pengcheng Jiang**, Cao Xiao, Minhao Jiang, Parminder Bhatia, Taha Kass-Hout, Jimeng Sun, and Jiawei Han, “Reasoning-Enhanced Healthcare Predictions with Knowledge Graph Community Retrieval”. [\[pdf\]](#)[\[code\]](#)
- [AAAI’25] **Pengcheng Jiang**, Cao Xiao, Tianfan Fu, Parminder Bhatia, Taha Kass-Hout, Jimeng Sun, and Jiawei Han, “Bi-level Contrastive Learning for Knowledge-Enhanced Molecule Representations”. [\[pdf\]](#)[\[code\]](#)
- [NeurIPS’24] **Pengcheng Jiang**, Lang Cao, Cao Xiao, Parminder Bhatia, Jimeng Sun, and Jiawei Han, “KG-FIT: Knowledge Graph Fine-Tuning Upon Open-World Knowledge”. [\[pdf\]](#)[\[code\]](#)
- [ICLR’24] **Pengcheng Jiang**, Cao Xiao, Adam Cross, and Jimeng Sun, “GraphCare: Enhancing Healthcare Predictions with Personalized Knowledge Graphs”. [\[pdf\]](#)[\[code\]](#)
- [NAACL’24 main] **Pengcheng Jiang**, Cao Xiao, Zifeng Wang, Parminder Bhatia, Jimeng Sun, and Jiawei Han, “TriSum: Learning Summarization Ability from Large Language Models”. [\[pdf\]](#)[\[code\]](#)
- [NAACL’24 main] **Pengcheng Jiang**, Jiacheng Lin, Zifeng Wang, Jimeng Sun, and Jiawei Han, “GenRES: Rethinking Evaluation for Generative Relation Extraction in the Era of LLMs”. [\[pdf\]](#)[\[code\]](#)
- [ACL’23 findings] **Pengcheng Jiang**, Shivam Agarwal, Bowen Jin, Xuan Wang, Jimeng Sun, and Jiawei Han, “Text-Augmented Open Knowledge Graph Completion via Pre-Trained Language Models”. [\[pdf\]](#)[\[code\]](#)
- [EMNLP’24 main] SeongKu Kang, Yunyi Zhang, **Pengcheng Jiang**, Dongha Lee, Jiawei Han, and Hwanjo Yu, “Taxonomy-guided Semantic Indexing for Academic Paper Search”. [\[pdf\]](#)[\[code\]](#)
- [KDD’23] Chaoqi Yang*, Zhengbang Wu*, **Patrick Jiang**, Zhen Lin, Junyi Gao, Benjamin Danek, and Jimeng Sun, “PyHealth: A Deep Learning Toolkit for Healthcare Predictive Modeling”. [\[pdf\]](#)[\[doc\]](#)[\[code\]](#)
- [IEEE SSCI’21] **Pengcheng Jiang**, Kenji Tei, “OACAL: Finding Module-consistent Specifications to Secure Systems from Weakened User Obligations”. [\[pdf\]](#)

USER STUDIES & TECH REPORTS

- [User Study] **Pengcheng Jiang**, Megan Lim, Adam Cross, and Jimeng Sun, “MedKG: Empowering Medical Education with Interactive Construction and Visualization of KGs via LLMs”, 2022. [\[pdf\]](#)[\[code\]](#)[\[software\]](#)

ACADEMIC EXPERIENCE

University of Illinois at Urbana-Champaign

Champaign, USA

Research Assistant to Prof. Jiawei Han and Prof. Jimeng Sun

Sep. 2022 – Present

- See “*Published Works*” and “*Preprints*” above.

Teaching Assistant (CS598: Deep Learning for Healthcare) to Prof. Jimeng Sun

Jan. 2023 – May 2023

- Provided one-on-one assistance to students on course material and programming assignments. Taught students the insights of popular deep learning (DL) models, and guided them to build their own pipelines of DL for health with PyHealth. Graded students’ project reports and presentations.

Teaching Assistant (CS101: Intro Computing) to Prof. Mattox Beckman

Aug. 2022 – Dec. 2022

- Led lab sections and office hours, designed over 200 questions for examinations, and taught fundamentals of data structures and algorithms, programming languages like Python and MATLAB to students.

Massachusetts Institute of Technology

Cambridge, USA

Research Projects with Prof. Mark Vogelsberger

May 2021 – Aug. 2021

- Developed a skin lesion diagnosis framework applying convolutional neural network models.

Waseda University

Tokyo, Japan

Research Assistant to Prof. Shinichi Honiden and Prof. Kenji Tei

Jun. 2020 – Jul. 2021

- Developed OACAL, an algorithm combining model checking and machine learning techniques to automatically revise software specifications

Teaching Assistant (Algorithms and Data Structures) to Prof. Honiden Shinichi

May 2020 – Sep. 2020

- Led lab sections and office hours teaching classic algorithms and data structures to students.

PROFESSIONAL EXPERIENCE

Google Research

Mountain View, USA

Student Researcher (Incoming)

June. 2025 – Aug. 2025

- Research information retrieval and reinforcement learning

GE HealthCare

Seattle, USA

AI Scientist Intern

May. 2024 – Aug. 2024

- Developed KARE (ICLR’25), which constructs a multi-sourced clinical concept knowledge graph and seamlessly integrates it with LLM reasoning for precise healthcare predictions.

Ant Group

Hangzhou, China

Software Engineer (AI Engineering)

Nov. 2021 – Jul. 2022

- Developed Auto-Refit, a tool embedded in the core MLOps for dynamic update of the deep learning models.
- Get engineering skills well-trained.

TALKS

- [2024.12] “Reasoning-Enhanced Healthcare Predictions with Knowledge Graph Community Retrieval”, invited by BioNLP Lab at UMass Amherst.

SERVICES

- **Paper Review:** NeurIPS, ICLR, ICML, EMNLP, NAACL, ACL, AAAI, AISTATS, KDD.